

ALFAAL07526

## 2-Mercaptobenzothiazole

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | 2-Mercaptobenzothiazole<br>2-Mercaptobenzothiazole                                                                                                                                                                                                                 |
| Cat No. :                     | L07526                                                                                                                                                                                                                                                             |
| Synonyms                      | 2-Benzothiazolethiol                                                                                                                                                                                                                                               |
| CAS No                        | 149-30-4                                                                                                                                                                                                                                                           |
| Molecular Formula             | C7 H5 N S2                                                                                                                                                                                                                                                         |
| Supplier                      | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                 |                      |                |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------|
| Physical State<br>Powder Solid                                                                                                  | Appearance<br>Yellow | Odor<br>Stench |
| <b>Emergency Overview</b><br>May cause an allergic skin reaction. Very toxic to aquatic life with long lasting effects. Stench. |                      |                |

#### Classification of the substance or mixture

|                          |            |
|--------------------------|------------|
| Skin Sensitization       | Category 1 |
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

#### Label Elements



Signal Word

Warning

## 2-Mercaptobenzothiazole

**Hazard Statements**

H317 - May cause an allergic skin reaction

H410 - Very toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P272 - Contaminated work clothing should not be allowed out of the workplace

P280 - Wear protective gloves

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

**Response**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P333 + P313 - If skin irritation or rash occurs: Get medical advice/attention

P362 + P364 - Take off contaminated clothing and wash it before reuse

**Storage**

P403 - Store in a well-ventilated place

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

May cause an allergic skin reaction.

**Environmental hazards**

Very toxic to aquatic life with long lasting effects. Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil. The product is insoluble and sinks in water.

**Other Hazards**

Stench. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component               | CAS No   | Weight % |
|-------------------------|----------|----------|
| 2-Mercaptobenzothiazole | 149-30-4 | 95       |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention.

**Inhalation**

Remove from exposure, lie down. Remove to fresh air. Get medical attention.

**Ingestion**

Clean mouth with water. Get medical attention.

**Most important symptoms and effects**

May cause allergic skin reaction. . Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Use personal protective equipment as required.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage.

**Methods for Containment and Clean Up**

Avoid dust formation. Sweep up and shovel into suitable containers for disposal. Do not let this chemical enter the environment.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Avoid contact with skin and eyes. Do not breathe dust. Do not ingest. If swallowed then seek immediate medical assistance. Handle product only in closed system or provide appropriate exhaust ventilation.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

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Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                 |                                 |                                          |
|---------------------------------|---------------------------------|------------------------------------------|
| <b>Appearance</b>               | Yellow                          |                                          |
| <b>Physical State</b>           | Powder Solid                    |                                          |
| <b>Odor</b>                     | Stench                          |                                          |
| <b>Odor Threshold</b>           | No data available               |                                          |
| <b>pH</b>                       | 7                               | 0.12 g/L (25°C)                          |
| <b>Melting Point/Range</b>      | 172 - 182 °C / 341.6 - 359.6 °F |                                          |
| <b>Softening Point</b>          | No data available               |                                          |
| <b>Boiling Point/Range</b>      | No information available        |                                          |
| <b>Flash Point</b>              | 243 °C / 469.4 °F               | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | Not applicable                  | Solid                                    |
| <b>Flammability (solid,gas)</b> | No information available        |                                          |
| <b>Explosion Limits</b>         | <b>Lower</b> 15 Vol%            |                                          |
| <b>Vapor Pressure</b>           | negligible                      |                                          |
| <b>Vapor Density</b>            | Not applicable                  | Solid                                    |

## 2-Mercaptobenzothiazole

|                                         |                          |                       |
|-----------------------------------------|--------------------------|-----------------------|
| Specific Gravity / Density              | 1.42                     |                       |
| Bulk Density                            | No data available        |                       |
| Water Solubility                        | 0.12 g/L (25°C)          | practically insoluble |
| Solubility in other solvents            | No information available |                       |
| Partition Coefficient (n-octanol/water) |                          |                       |
| Component                               | log Pow                  |                       |
| 2-Mercaptobenzothiazole                 | 2.42                     |                       |
| Autoignition Temperature                | 465 °C / 869 °F          |                       |
| Decomposition Temperature               | > 260°C                  |                       |
| Viscosity                               | Not applicable           | Solid                 |
| Explosive Properties                    | No information available |                       |
| Oxidizing Properties                    | No information available |                       |

|                   |            |
|-------------------|------------|
| Molecular Formula | C7 H5 N S2 |
| Molecular Weight  | 167.24     |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                              |
|--------------------------|----------------------------------------------|
| Stability                | Stable under normal conditions.              |
| Hazardous Reactions      | No information available.                    |
| Hazardous Polymerization | No information available.                    |
| Conditions to Avoid      | Avoid dust formation. Incompatible products. |
| Materials to avoid       | Strong oxidizing agents.                     |

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component               | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation             |
|-------------------------|---------------------------|------------------------------|-----------------------------|
| 2-Mercaptobenzothiazole | LD50 = 3800 mg/kg ( Rat ) | LD50 > 7940 mg/kg ( Rabbit ) | LC50 > 0.7 mg/L ( Rat ) 7 h |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

## (d) respiratory or skin sensitization;

Respiratory  
Skin

Based on available data, the classification criteria are not met  
Category 1

May cause sensitization by skin contact

## (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met  
Not mutagenic in AMES Test

## (f) carcinogenicity;

Based on available data, the classification criteria are not met  
The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component               | EU | UK | Germany | IARC     |
|-------------------------|----|----|---------|----------|
| 2-Mercaptobenzothiazole |    |    |         | Group 2A |

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|                                            |                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (g) reproductive toxicity;                 | Based on available data, the classification criteria are not met                                                                                                                      |
| (h) STOT-single exposure;                  | Based on available data, the classification criteria are not met                                                                                                                      |
| (i) STOT-repeated exposure;                | Based on available data, the classification criteria are not met                                                                                                                      |
| Target Organs                              | None known.                                                                                                                                                                           |
| (j) aspiration hazard;                     | Not applicable<br>Solid                                                                                                                                                               |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information                                                                     |
| Symptoms / effects, both acute and delayed | Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |

## SECTION 12. ECOLOGICAL INFORMATION

|                     |                                                                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ecotoxicity effects | Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Component               | Freshwater Fish                                                                                                                                                      | Water Flea                            | Freshwater Algae                                         | Microtox            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------|---------------------|
| 2-Mercaptobenzothiazole | LC50: = 11 mg/L, 96h static (Pimephales promelas)<br>LC50: = 0.42 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: 1.32 - 2.73 mg/L, 96h static (Lepomis macrochirus) | EC50: = 4.1 mg/L, 48h (Daphnia magna) | EC50: = 0.25 mg/L, 96h (Pseudokirchneriella subcapitata) | EC50 = 10 mg/L 24 h |

|                                       |                                                                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Persistence and Degradability         | Not readily biodegradable                                                                                       |
| Persistence                           | Insoluble in water, Persistence is unlikely.                                                                    |
| Degradation in sewage treatment plant | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. |

|                           |                                          |
|---------------------------|------------------------------------------|
| Bioaccumulative Potential | May have some potential to bioaccumulate |
|---------------------------|------------------------------------------|

| Component               | log Pow | Bioconcentration factor (BCF) |
|-------------------------|---------|-------------------------------|
| 2-Mercaptobenzothiazole | 2.42    | <0.8 dimensionless            |

|                  |                                                                                                                                                      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mobility in soil | Spillage unlikely to penetrate soil The product is insoluble and sinks in water Is not likely mobile in the environment due its low water solubility |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                 |                                                                           |
|---------------------------------|---------------------------------------------------------------------------|
| Endocrine Disruptor Information | This product does not contain any known or suspected endocrine disruptors |
| Persistent Organic Pollutant    | This product does not contain any known or suspected substance            |
| Ozone Depletion Potential       | This product does not contain any known or suspected substance            |

## SECTION 13. DISPOSAL CONSIDERATIONS

## 2-Mercaptobenzothiazole

|                                            |                                                                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                           |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.                |

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

|                                |                                                     |
|--------------------------------|-----------------------------------------------------|
| <b>UN-No</b>                   | UN3077                                              |
| <b>Proper Shipping Name</b>    | Environmentally hazardous substances, solid, n.o.s. |
| <b>Technical Shipping Name</b> | 2-Mercaptobenzothiazole                             |
| <b>Hazard Class</b>            | 9                                                   |
| <b>Packing Group</b>           | III                                                 |

IMDG/IMO

|                                |                                                     |
|--------------------------------|-----------------------------------------------------|
| <b>UN-No</b>                   | UN3077                                              |
| <b>Proper Shipping Name</b>    | Environmentally hazardous substances, solid, n.o.s. |
| <b>Technical Shipping Name</b> | 2-Mercaptobenzothiazole                             |
| <b>Hazard Class</b>            | 9                                                   |
| <b>Packing Group</b>           | III                                                 |

IATA

|                                |                                                     |
|--------------------------------|-----------------------------------------------------|
| <b>UN-No</b>                   | UN3077                                              |
| <b>Proper Shipping Name</b>    | Environmentally hazardous substances, solid, n.o.s. |
| <b>Technical Shipping Name</b> | 2-Mercaptobenzothiazole                             |
| <b>Hazard Class</b>            | 9                                                   |
| <b>Packing Group</b>           | III                                                 |

**Special Precautions for User** No special precautions required

## SECTION 15. REGULATORY INFORMATION

**International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component               | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| 2-Mercaptobenzothiazole | -                                                   | -                                       | X    | X     | 205-736-8 | X    | X   | X     | X    | X    | X    | KE-02723 |

**National Regulations**

## SECTION 16. OTHER INFORMATION

## 2-Mercaptobenzothiazole

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 07-Jun-2010  
**Revision Date** 27-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

First aid for chemical exposure, including the use of eye wash and safety showers.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

Chemical incident response training.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**